

Mechanical Characterization of a Surgical Endo pouch: Assessing Burst Strength, Puncture Resistance and Insertion/Extraction Durability

Name	Student Number	E-mail
Tony Euwals	5679400	Tonyeuwals@tudelft.nl
Elvira Zonneveld	6008038	ezonneveld@tudelft.nl
Geert Wormgoor	6573843	gwormgoor@tudelft.nl
Elliot Czarik	6574165	S3966879@vuw.leidenuniv.nl
Jibiao Zhang	5978882	J.Zhang-94@student.tudelft.nl

Abstract

During a morcellation procedure in minimally invasive laparoscopic surgery a tissue containment system, such as an endo pouch, is required to reduce the risk of spread of malignant cells which could result in metastasis. If the integrity of the endo pouch is affected in any way during the procedure there will be spillage of the cells contained which could still result in spread of malignant cells. This study developed four tests to assess the mechanical safety and integrity of a newly designed endo pouch during four critical phases of its use.

The insertion test measured forces required to pull an empty pouch through a trocar into the abdominal cavity. The trocar sizes were 8 to 15 mm sizes with ranging results from 71.4 ± 9.43 N for the 10 mm trocar size to 6.69 ± 3.29 N for the 15 mm trocar. The extraction test quantified the required amount of force to extract an endo pouch post morcellation procedure with varying amounts of residual tissue and incision sizes. The amount of tissue varied from 5 to 35 grams with incision sizes varying from 10 to 24 mm. Results showed holes larger than 22 mm required minimal force and holes smaller than 16 mm exhibited failures with forces larger than 32 ± 4.9 N. No leakage or endo pouch failure occurred during testing. Puncture and burst tests were designed with protocol but not executed due to time and equipment limitations.

Keywords: *endo pouch, laparoscopic surgery, morcellation, tissue containment system, mechanical testing, medical device*

1. Introduction

Laparoscopic surgeries are minimally invasive procedures during which the abdomen or pelvis is accessed via small incisions. Specific examples include myomectomies (an operation to remove fibroids) and hysterectomies (an operation to remove the uterus). To remove large tissue via small incisions, a round rotating blade breaks the specimen into smaller fragments via morcellation. This approach is

accompanied with a risk of spreading cancerous cells, significantly reducing the patient's long-term survival. Therefore, these procedures are performed within tissue containment systems (endo pouches), which provide an isolated workspace. Its primary function is to serve as a barrier between target tissue and surrounding organs. [1-4]

To insert the bag into the abdominal cavity, it is pulled through a trocar using a Maryland clamp. The bag is then insufflated with CO₂ to a pressure of 15 mmHg, providing a

workspace for tissue manipulation. Using a grasper, the target tissue is placed into the bag, followed by morcellation of the tissue. The surgeon may then extract large individual residual tissue with the grasper, until the bag is assessed empty enough for extraction. The bag can then be extracted directly through a skin incision, with minimal residual tissue.

Despite the range of endo pouches currently available, several challenges remain regarding their sustainability and ease of application, according to Dr Johann Rhemrev. A new design, see figure 1.1, aims to tackle these issues, whilst providing the same level of safety as existing pouches. For instance, by introducing an additional port, an extra maryland clasper is available for tissue manipulation, eliminating spinning effects of the specimen during morcellation. A further notable improvement includes reduced use of disposable assisting instruments. By decreasing overall surgery time and minimizing material usage, this prototype can significantly reduce CO₂ emissions associated with morcellation procedures.



Figure 1.1: The new endo pouch design

To evaluate the safety characteristics of the pouch, we propose four distinct tests, each designed to assess a specific hazard associated with its use. In 2021, a paper was published reporting on similar objectives, focussing on a similar prototype. The paper concluded that the bag could enhance safety of in-bag morcellation, and may potentially shorten surgery time. [5]

1.1 Problem statement

Potential failures during endo pouch application include tearing, puncturing, and rupturing of the endo pouch as shown in table 1.1. These can result in loss of containment,

posing a safety hazard, and increase of surgery time, compromising sustainability. The following table summarises potential risks during specific phases of endo pouch application, and the proposed corresponding test.

Table 1.1: Process phase and their corresponding potential failure

PROCESS PHASE	POTENTIAL CAUSE	FAILURE MODE	FAILURE EFFECT	TEST
EXTRACTION	Applying excessive force	Bag tearing	Loss of containment	Extraction test
INSERTION	Applying excessive force	Bag tearing	Loss of integrity	Insertion test
MANIPULATION OF BAG & TISSUE	Contact with sharp instrument	Bag puncture	Potential leakage and loss of integrity	Puncture test
INSUFFLATION	Over-insufflation of the bag	Bag rupture	Loss of containment and loss of integrity	Burst test

1.2 Research question / Aim

This study aims to develop and describe four distinct reproducible tests which assess the safety characteristics of an endo pouch. These tests evaluate the following: puncture force, insertion force, extraction force, and whether the bag can withstand clinically relevant pressures.

The extraction and insertion test were conducted on the newly designed endo pouch to characterise its mechanical properties and to assess its durability. Accordingly, two research questions are formulated:

- I) I. "For different clinically relevant incision sizes, how much force is required to extract the endo pouch for varying masses of contained tissue?"
- II) "For different clinically relevant trocar sizes, how much force is required to insert the endo pouch?"

1.3 Relevance

Rigorous and correct testing of the endo pouch minimises the risk of failure during surgery, supporting surgeons in providing the safest possible treatment. By mechanically characterising the prototype, potential failure sites can be detected and corrected. Furthermore, improving endo pouch sustainability contributes to environmentally responsible surgical practice.

2. Materials & methods: Extraction Test

2.1 Materials

An extraction board (see figure 2.1), fabricated from polymethylmethacrylate (PMMA), was used to simulate the extraction incisions through which the bag is removed. PMMA was selected based on its availability. The board features holes with diameters ranging from 3mm to 24mm, with relevant sizes starting at 10mm. Commonly utilised clinical trocars for in-bag morcellation have diameters of 12 or 15 mm [6,7], although the corresponding incisions vary.

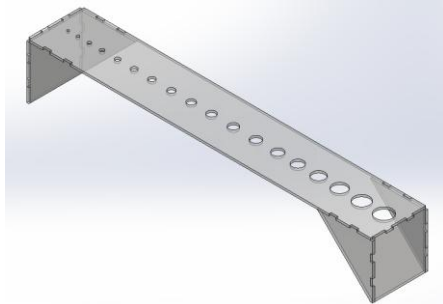


Figure 2.1: Extraction board displayed in SolidWorks

To mimic the residual tissue, we used chicken fillet due to its availability and relatively close resemblance to clinically relevant tissue. Additionally, blue food dye was used for leakage detection as this provides a high contrast. For quantifying the forces, a portable electronic scale [8] was employed. The following formula was used to calculate force:

$$F = m \cdot g \quad (1)$$

With gravitational acceleration $g \approx 9.81m/s^2$, force F in Newtons and mass m in kilograms.

2.2 Procedure

35g of chicken fillet was cut into clinically relevant sizes. Assessing whether a piece is appropriately sized requires prior morcellation experience, as clinical sizing largely depends on the surgeon's judgement. We added blue food dye to the chicken, which was then divided into 5g portions. To prepare the bag for testing it was sealed at the designated sealing sites (this prototype has two). The specified tissue mass was then placed into the bag via the big opening. To ensure a minimal amount of air is trapped within the bag, we smoothed it by hand, after which we loosely sealed it with a zip tie. A second zip tie was secured around the first, providing a hooking site for the electronic scale. To ensure the board remained stationary, the board was clamped with a vice. We then pulled the bag through the extraction board using the electronic scale, with the applied force oriented perpendicular to the board.

We tested all masses (5, 10, 15, 20, 25, 30 and 35g) across all diameters (10, 11, 12, 13, 14, 15, 16, 18, 20, 22 and 24 mm). Each trial was repeated 5 times. Between each trial the bag was visually inspected for any leakage and shaken to redistribute the contained tissue. To ensure methodological consistency, the same operator should perform each measurement.

3. Materials & methods: Insertion Test

3.1 Materials

An insertion board (see figure 3.1), fabricated from Acrylonitrile Butadiene Styrene (ABS) using 3D printing, served to simulate trocar insertion. We selected ABS because it replicates the plastic material used for clinical trocars [9]. The board features removable tubes of varying diameters of 8, 10, 11, 12, 15 mm, of which 12 and 15 mm reflect true trocar sizes. [6,7]

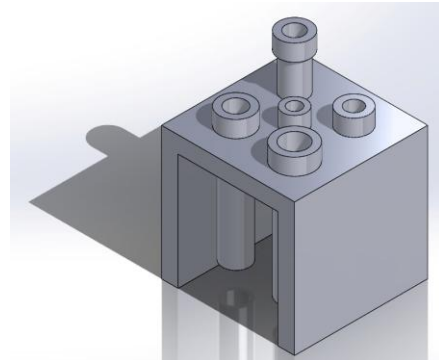


Figure 3.1: Insertion board displayed in SolidWorks

Similar to the extraction test, the same electronic scale and formula (1) was utilised to quantify the applied forces.

3.2 Procedure

We attached the testing tube to the electronic scale with duct tape (figure 3.2(a)). The handle of the Maryland clasper was clamped in a vice, whereby the handles were still operational (figure 3.2(b)). We rolled the bag as tightly as possible, resembling its surgical use. One end of the bag was then secured by the Maryland clamp, and the tube was placed on top with the scale suspended downwards. Whilst filming, we carefully pulled the tube downwards, and the maximum weight was recorded (figure 3.2(c)). Five trials were conducted for each tube, with the bag rerolled and the scale newly calibrated in between each trial.

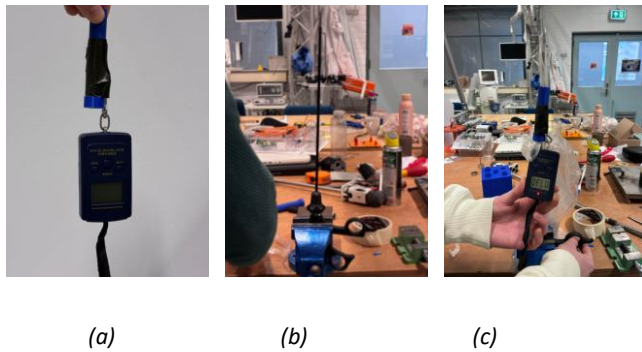


Figure 3.2: (a) The electronic scale attached to the testing tube, (b) the Maryland clasper fastened in a vice, (c) screenshot of the video made during a measurement.

4. Materials & methods: Puncture test

4.1 Materials

The puncture test requires 15 x 15 cm material specimen for specified thicknesses. To clamp the specimen, we designed a specimen holder, displayed in figure 4.1, the specimen sheet can be placed between the two rings and clamped using the red clamps. This specimen holder also features an additional alignment guide, illustrated with a yellow colour in the figure. The alignment guide ensures force is applied perpendicular to the clamped specimen.

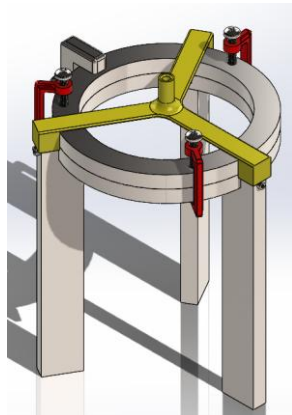


Figure 4.1: 3D model of the specimen holder displayed in SolidWorks

Additional materials required include a kitchen scale to measure the force applied, and a Maryland clamp (or other clinically relevant surgical instruments) to apply the puncture force.

4.2 Procedure

To verify the thickness of each material specimen, a calliper can be used. For accurate evaluation, measurements should be repeated five times at different regions of the

sheet. The specimen can then be secured in the clamping fixture, such that the sheet is centrally aligned. Ensure the sheet is sufficiently tensioned to eliminate wrinkles. The C-clamps should be employed as shown in figure 4.2. The clamping fixture with the specimen installed is then placed onto the kitchen scale, which is subsequently calibrated. To apply force, the Maryland clamp is inserted into the alignment guide and pushed downwards into the specimen. The force should gradually increase in a linear manner, until specimen failure (e.g. puncture). The maximum weight displayed on the kitchen scale is noted down and converted into newtons using equation (1).

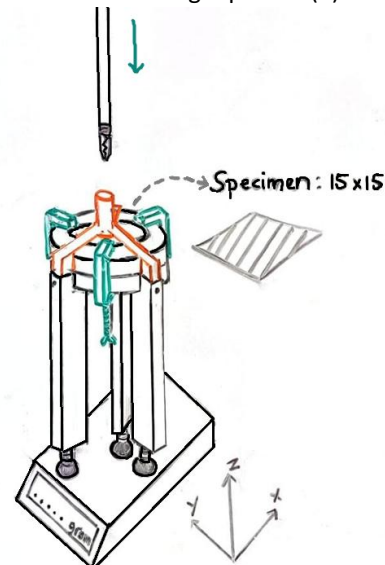


Figure 4.2: full test setup of the puncture test

5. Materials & methods: Burst test

5.1 Materials

To insufflate the pouch, an inflation device is required with accurate pressure control and an airtight connection to the pouch. The instrument must provide pressures up to 50 mm Hg at a minimum. For simplicity, we suggest using a modified blood pressure cuff equipped with an mmHg pressure gauge. Medical insufflators can also deliver low pressures in the mmHg range, but most are limited to a maximum of about 25 mmHg as a clinical safety precaution. Therefore, they are not suitable when higher pressures are required. Further equipment includes a large transparent, sealable plastic container, blue food colouring and suitable hearing protection.

5.2 Procedure

Seal one of the smaller pouch openings using the inward double knot method that was researched for the Insamedso

endo pouches a few years ago.[10] Connect the other small opening to the inflation device. Mix 20 ml of water with sufficient blue food colouring and pour into the bag via the big opening (with a funnel). Seal the big opening of the bag in an airtight manner, we suggest using a knot and a zip tie. Place the bag into the transparent plastic container, resulting in a test setup as shown in figure 5.1.

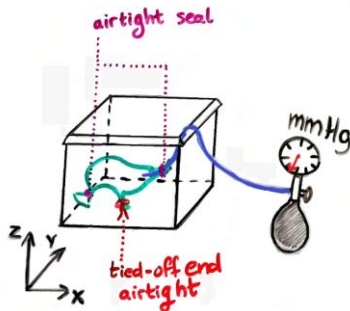


Figure 5.1: Full test setup of the burst test.

Whilst filming, apply an internal pressure of 15 mmHg to the pouch and hold this system for 10 minutes to verify the bag is sealed airtight. Increase the pressure by 5 mmHg increments, with a 3-minute stabilization and observation period between each raise in pressure. Test the pouch until failure (e.g. burst). This method is inspired by ASTM F2054 for package burst testing [11].

6. Results

This chapter will be subdivided into insertion test and extraction test, since only these two tests were successfully performed.

Extraction test:

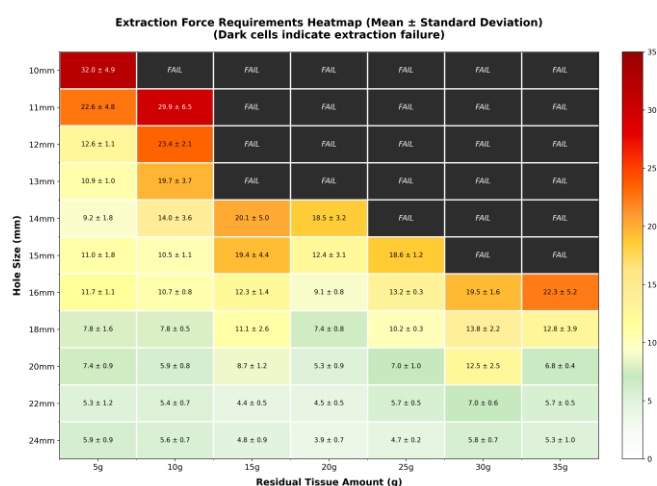


Figure 6.1: Force required for extraction in Newtons for every residual tissue amount in grams in combination with hole size in millimeters.

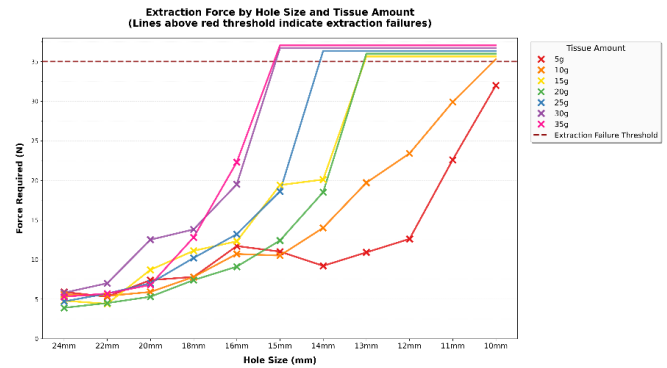


Figure 6.2: Line through points of required force in Newtons against hole size in millimeters for every residual tissue amount in grams.

The heatmap (Figure 6.1) and line chart (Figure 6.2) demonstrate extraction force requirements across hole sizes (10-24mm) and tissue amounts (5-35g). Three distinct performance zones were identified: hole sizes 22-24mm required minimal force (3.9-7.0N) with 100% success across all tissue amounts; the 16-20mm range showed moderate forces (5.3-22.3N) with complete success; and sizes below 16mm exhibited progressive extraction failures marked by dark cells in the heatmap and lines above the 35N threshold in the line chart (Figure 6.2).

The data suggests a clear inverse relationship between hole size and extraction force. Smaller holes (<16mm) demonstrated both higher forces and earlier failures as tissue amount increased, with the 10mm hole failing at tissue amounts $\geq 10g$ despite requiring the maximum recorded force of $32.0 \pm 4.9N$ at 5g. Conversely, larger holes ($\geq 18mm$) maintained stable, low force requirements regardless of tissue burden, indicating that adequate size effectively mitigates tissue-related mechanical resistance.

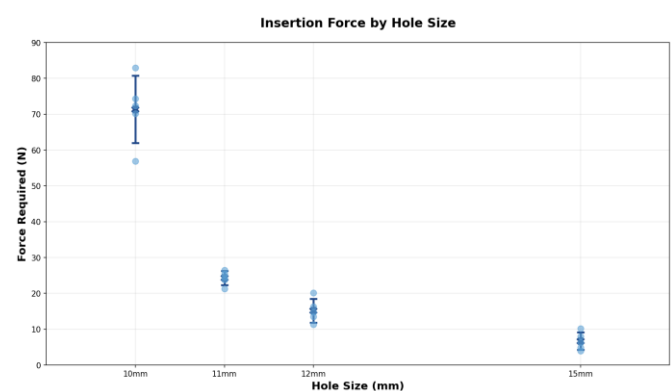


Figure 6.3: Required force for insertion in Newtons against hole size in millimeters.

Insertion test:

Figure 6.3 demonstrates insertion force requirements across trocar sizes ranging from 10mm to 15mm. The data reveal a strong inverse relationship between trocar diameter and insertion force, with mean forces of $71.4 \pm 9.43\text{N}$ (10mm), $24.4 \pm 1.95\text{N}$ (11mm), $15.19 \pm 3.29\text{N}$ (12mm), and $6.69 \pm 3.29\text{N}$ (15mm).

The 10mm trocar required approximately 8-fold greater insertion force compared to the 15mm trocar, representing the most substantial performance differential observed. Additionally, the 8mm configuration we tested reached forces beyond failure and even got stuck, suggesting less predictable insertion mechanics at smaller diameters. The progressive reduction in both mean force and variability with increasing trocar size indicates that larger diameter instruments provide more consistent and lower-resistance tissue penetration.

These findings suggest that trocar sizes $\geq 12\text{mm}$ offer substantially reduced insertion forces with improved procedural consistency compared to smaller diameter options.

7. Discussion

7.1 Extraction test

The extraction test indicated that the pouch could withstand the use of excessive force, as no leakage was detected. Results suggest that for masses up 35g, incisions of 18 mm and larger allow for a smooth extraction process, whilst smaller holes offer substantial resistance. Based on these findings, a minimum hole size of 18mm is recommended for reliable extraction, while 22-24mm provides optimal performance with the lowest forces and complete procedural success.

A major limitation arises from the use of chicken fillet to mimic the residual morcellation tissue. Tissue commonly subjected to morcellation, such as the uterus or myomas, are relatively stiff.[12] Chicken fillet in comparison, is much softer. As a result, the data may underestimate the force required to extract the pouch. Furthermore, unlike our test board, skin is flexible and thus expands if force is applied. Additionally, in clinical circumstance the pouch is stained with fresh blood, which can act as a lubricant during extraction and reduce friction effects [13]. For further research, we suggest using cow tongue instead, and applying a silicone cover over the test board, as shown in figure 7.1, to mimic the skin.[14]

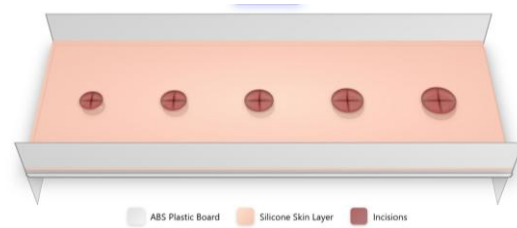


Figure 7.1: The extraction board simulated with a silicone skin layer

We did not manage to perform the entire extraction test with one operator, introducing an additional source of error.

7.2 Insertion test

The insertion test demonstrated that the endo pouch requires significantly less force for insertion as trocar diameter increases, with a particularly strong inverse relationship observed across the tested range of 10-15mm. This substantial force differential suggests that trocar selection has critical implications for both procedural ease and potential tissue trauma during insertion. The data strongly support the use of trocars $\geq 12\text{mm}$, which demonstrated forces below 16N with relatively low variability ($\pm 3.29\text{N}$), offering a favourable balance between mechanical efficiency and procedural reliability.

Notably, the 8mm configuration experienced forces beyond the measurement threshold and became stuck during testing, suggesting a critical lower limit exists below which insertion becomes mechanically unfavourable and unpredictable. This finding aligns with clinical practice guidelines that typically recommend minimum trocar sizes of 10-12mm for practical use of this prototype.

Our biggest inconsistency arose from the fact that the bags were rolled manually to simulate clinical practice, introducing operator-dependent variability in bag compactness and uniformity. Tighter rolling would theoretically reduce the effective diameter of the rolled bag, potentially lowering insertion forces, while looser rolling could increase resistance. Although efforts were made to standardize the rolling technique, this remains a source of potential inconsistency. Like the extraction test, it was unfortunately also not possible to perform the entire extraction test with one operator, introducing an additional source of error.

7.3 Puncture test

The puncture test was designed to be low cost and simple, with readily available materials. Therefore, we proposed that

the force should be applied manually, which would introduce substantial deviations in the rate of force applied between each trial. Thus, we suggest using a linear stage when possible as an alternative.

For the PMU specimens, we found differing opinions in the literature regarding the minimum required size. Some studies suggest that a 5 × 5 cm specimen is sufficient, while others report that edge effects may be noticeable at this scale, potentially influencing the test results. [15,16] To minimize these potential influences, we therefore opted for a larger specimen size of 15 × 15 cm. This size is compatible with our clamping fixture, which has an inner diameter of 10 cm, ensuring proper fixation during testing. Although a smaller specimen size could be used, it is important to consider the potential disadvantages associated with this choice.

7.4 Burst test

The main challenge of the burst test protocol is the need for a pump capable of reaching pressures beyond those of the available clinical insufflator, while still operating accurately in the mmHg range. Most standard air pump devices, such as compressors (typically 0–8 bar, accuracy ±0.5 bar[17]) or bicycle pumps (typically 0–11 bar, with decreasing accuracy at lower pressures[18]), are unsuitable because they cannot reliably generate or measure pressures in the required range of 0–100 mmHg.

Although specialized ultra-low- pressure pumps exist, these devices are highly specialized and typically cost between €2000 and €3000.[19] Due to these constraints, the idea of modifying a blood pressure cuff was explored. Blood pressure cuffs operate in a pressure range of approximately 0–300 mmHg, making them theoretically suitable for this application.[20] However, the feasibility of this approach would require further testing.

Several technical challenges were identified with this solution:

Firstly, it would be necessary to create an airtight connection between the blood pressure device and the Endo pouch without restricting airflow, as any leakage would make it difficult to maintain a constant internal pressure.

Secondly, since the blood pressure cuff uses a manual pump bulb, achieving and maintaining a stable pressure plateau—required for controlled testing—would be difficult.

Thirdly, this device wouldn't work with CO₂ like the clinical insufflator but works with O₂. Since CO₂ and O₂ have a similar composition [21], the difference in gas used is not expected to make a significant difference in this test.

Ideally, we would have an insufflator that supports higher pressures. In that case most of these design challenges would be dealt with, as the insufflator can keep the pressure stable

even without a fully air-tight seal. We would be able to use CO₂-gas as they do clinically and follow our protocol to start from 15 mmHg (clinical used pressure) and stepwise increase the pressure with increments of 5 mmHg with 3-minute observation times in between to check for leaks.

Due to patient safety, all medical insufflators have a maximum pressure of 25 mmHg. Exceeding this pressure would increase hemodynamic and respiratory risks during laparoscopic surgery and is advised against. [22]

8. Conclusion

Based on the results of the insertion and extraction tests, it can be concluded that the endo pouches exhibit sufficient tensile strength for practical use. In clinical setting, this means the pouch is expected to be capable of handling all forces experienced during the insertion and extraction phase, if handled correctly.

For other parameters, such as puncture resistance, strength of the pouch when inflated, and seam strength, additional testing is required. These parameters are the primary focus of the other tests we have designed: the burst test and the puncture test.

Acknowledgements

We would like to express our gratitude to Dr. Johann Rhemrev and Dr. Jim English for their guidance and expertise throughout this research project. We are particularly grateful to Dr. Johann Rhemrev for generously providing the materials necessary to conduct the experiments.

We also want to extend our appreciation to Prof.dr. John van den Dobbelsteen for his advice to the development of the testing methodologies.

References

- [1] National Health Service (NHS), "Laparoscopy," NHS, U.K. [Online]. Available: <https://www.nhs.uk/tests-and-treatments/laparoscopy/>. Accessed: Nov. 2025.
- [2] Royal College of Obstetricians and Gynaecologists (RCOG), "Morcellation for myomectomy or hysterectomy," RCOG, London, U.K. [Online]. Available: <https://www.rcog.org.uk/for-the-public/browse-our-patient-information/morcellation-for-myomectomy-or-hysterectomy/>. Accessed: Nov. 2025.
- [3] U.S. Food and Drug Administration, "Laparoscopic power morcellators," FDA, Silver Spring, MD, USA. [Online]. Available: <https://www.fda.gov/medical-devices/surgery-devices/laparoscopic-power-morcellators>. Accessed: Nov. 2025
- [4] M. Einarsson *et al.*, "Contained power morcellation within an insufflated isolation bag," *J. Minim. Invasive Gynecol.*, vol. 17, no. 6, pp. 776–781, 2010. Available: <https://www.sciencedirect.com/science/article/pii/S1553465010002244?via%3Dihub>
- [5] S. Cohen *et al.*, "Evaluation of containment systems for laparoscopic power morcellation," *Sci. Rep.*, vol. 11, Art. no. 19993, 2021. Available: <https://www.nature.com/articles/s41598-021-99934-1>
- [6] A. Tinelli *et al.*, "Laparoscopic myomectomy with contained morcellation," *Int. J. Gynecol. Obstet.*, vol. 2017, Art. ID 9410571, 2017. Available: <https://onlinelibrary.wiley.com/doi/10.1155/2017/9410571#:~:text=Usually%20we%20realize%20this%20procedure,pr%20eserves%20the%20minimally%20invasive%20approach>
- [7] J. B. McGreevy *et al.*, "Trocarr site hernia following laparoscopic surgery," *JSLs*, vol. 15, no. 3, pp. 343–347, 2011. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC3134687/#:~:text=in%20hernia%20development,-CONCLUSION,or%20widened%20the%20initial%20defect>[11] https://store.astm.org/f2054_f2054m-13r20.html
- [8] Digitale kofferweegschaal, bol.com, The Netherlands. [Online]. Available: <https://www.bol.com/nl/nl/p/digitale-kofferweegschaal-bagage-weegschaal/9200000051252471/>. Accessed: Dec. 2025.
- [9] Manners Medical, "Disposable trocar materials," Manners Medical, U.K. [Online]. Available: <https://www.mannersmedical.com/industries/67.html>. Accessed: Dec. 2025.
- [10] A. Hilker, L. Schwander, M. Looye, and T. Kaal, "The use of containment bags during laparoscopic procedures: A research in which a containment bag prototype is tested in order to determine its safety when used for hysterectomies and myomectomies," Research report (unpublished), Delft University of Technology, Delft, The Netherlands, 2020.
- [11] ASTM International, *ASTM F2054/F2054M-13(2020)*, Standard Test Method for Measurement of Magnetically Induced Displacement Force, West Conshohocken, PA, USA, 2020. Available: https://store.astm.org/f2054_f2054m-13r20.html
- [12] J. A. Stewart *et al.*, "Uterine fibroids: Pathophysiology and clinical features," *Obstet. Gynecol. Int.*, vol. 2014, Art. ID 321685, 2014. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC4106177/#:~:text=Uterine%20fibroids%20are%20firm%2C%20stiff,to%20activation%20of%20signaling%20cascades>[13] <https://www.sciencedirect.com/science/article/pii/S2405451816300472>
- [13] S. Zhang *et al.*, "Tribological behavior of biomedical polymers," *Biosurface and Biotribology*, vol. 2, no. 4, pp. 173–181, 2016. Available: <https://www.sciencedirect.com/science/article/pii/S2405451816300472>
- [14] A. Ramírez-Pérez *et al.*, "Skin friction and lubrication mechanisms," *Skin Res. Technol.*, vol. 21, no. 4, pp. 394–400, 2015. Available: <https://onlinelibrary.wiley.com/doi/10.1111/srt.12235>[17] https://www.brannan.co.uk/knowledge_base/pressure-gauges-accuracy-explained/
- [15] Universal Grip Co., "ASTM F1306 testing," Universal Grip Co., USA. [Online]. Available: <https://www.universalgripco.com/astm-f1306>. Accessed: Dec. 2025.
- [16] AHP Makina, "ASTM D4833 puncture resistance test equipment," AHP Makina, Turkey. [Online]. Available: <https://ahp-makina.com/astm-d-4833-standard-test-method-for-index-puncture-resistance-of-geomembranes-and-related-products-testing-equipment/>. Accessed: Dec. 2025.
- [17] Brannan, "Pressure gauges accuracy explained," Brannan, U.K. [Online]. Available: https://www.brannan.co.uk/knowledge_base/pressure-gauges-accuracy-explained/. Accessed: Dec. 2025.

[18] Outrider Components, “How accurate are pump pressure gauges?” Outrider Components, USA. [Online]. Available:
<https://www.outridercomponents.com/blogs/news/how-accurate-are-pump-pressure-gauges>. Accessed: Dec. 2025.

[19] Chamois Inc., “Ultra-low pressure calibration pumps,” Chamois Inc., USA. [Online]. Available:
<https://www.chamois.net/products/fluke/ultra-low-pressure-calibration-pumps/>. Accessed: Dec. 2025.

[20] bol.com, “Romed aneroid blood pressure meter,” bol.com, The Netherlands. [Online]. Available:
<https://www.bol.com/nl/nl/p/romed-bloeddrukmeter-aneroid-palm-type-handmatige-bloeddrukmeter/9300000081315534/>. Accessed: Dec. 2025.

[21] J. Wang *et al.*, “Thermophysical properties of CO₂–O₂ mixtures,” *J. Chem. Eng. Data*, vol. 68, no. 11, pp. 3521–3530, 2023. Available:
<https://pubs.acs.org/doi/10.1021/acs.jced.3c00131>

[22] DrOracle AI, “Recommended maximum intra-abdominal pressure for carbon dioxide insufflation,” DrOracle, USA. [Online]. Available:
<https://www.droracle.ai/articles/482160/what-is-the-recommended-maximum-intra-abdominal-pressure-for-carbon>. Accessed: Jan. 2026.

Appendix A: Photos during tests



Figure A.1: Morcellation of a cow's tongue under supervision of Johann. It was very helpful to see how morcellation works and what residue will be left after this procedure.



Figure A.2: Extraction test we performed on our first test day. The endo pouch is pulled through a hole in the PMMA extraction board with the help of two zip ties. For our second test day, we replaced this analogue luggage scale for a digital variant and were also more aware how to deal with leftover air in the bag.